KNOWING WHEN NOT TO TRADE: PHASE 2 TECHNICAL REPORT

Uncertainty-Aware Price Forecasting to Protect Debt-Financed Vegetable Traders in Sri Lanka

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1. Problem Grounding: The Microfinance Downside Trap

Small vegetable traders in Sri Lanka borrow daily working capital before dawn from informal lenders at high interest to buy perishable produce at the **Dambulla Dedicated Economic Centre (DDEC)** at wholesale price P_t^w and truck it to **Colombo terminal markets (Peliyagoda/Pettah)** to sell the next day at P_{t+1}^r. For trip interest rate r and transport/spoilage cost C, profit and break-even are:

$$\Pi_t = P_{t+1}^r - P_t^w - C - r\,P_t^w \qquad \Longleftrightarrow \qquad \text{Break-Even Price: } B_t = P_t^w(1+r) + C \tag{1}$$

While lenders collect principal and interest unconditionally, the trader carries **100% of downside price risk**. Conventional ML models output single-point estimates (R^2, MAE) that convey false certainty: when sudden gluts crash urban prices, point forecasts encourage ruinous borrowing, driving rural families into debt bondage. **Model Y replaces point forecasting with calibrated 90% prediction intervals ($[L_t, U_t]$), Adaptive Conformal Inference (ACI), and a 3-tier protective decision rule:**

- **Green (OK to Trade):** 5th-percentile price $L_t \geq B_t$. Even a severe 5th-percentile bad day covers loan interest and transport. Full quota.
- **Amber (Buy Less / Own Cash):** Median $M_t \geq B_t > L_t$. Expected trade is profitable, but downside loss is plausible. Half quota, own capital.
- **Red (Do Not Borrow):** $M_t < B_t$, or 90-day route margin negative, or data stale (>4 days). **Price forecast is hidden to prevent anchoring bias.**

2. System Architecture & Technical Stack Rationale

- **Tabular Gradient Boosting (LightGBM):** Chosen over deep architectures (e.g. TFT) because tree ensembles train in seconds on economic time series, handle missing values natively, and optimize asymmetric pinball loss directly at quantiles $\tau \in \{0.05, 0.50, 0.95\}$.
- **CQR + Adaptive Conformal Inference (ACI):** Gibbs & Candès (2021) ACI ($\gamma = 0.02$) updates nominal error rate $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \mathbf{1}\{y_t \notin C_t\})$ on a 90-day rolling window. Validated against MAPIE 1.6 with exact numerical agreement (diff $= 0.0$).
- **Ratio Forecasting (Overcoming Tree Extrapolation Limits):** Trees cannot extrapolate beyond historical price ranges. By forecasting the spatial ratio $y_t = P_{t+1}^r / P_t^w$, the model generalizes seamlessly across inflationary regimes (e.g. 2022 diesel and price surges).
- **Client-Side PWA & Automated GitHub Actions Pipeline:** Implemented as an offline-first Progressive Web App (PWA) with local browser storage so traders' cost parameters never leave their device. Daily data ingestion runs via GitHub Actions at 06:00 and 10:00 SL time.

End-to-End Pipeline:

HARTI & CBSL PDFs ⟶ PyMuPDF / Regex Parser ⟶ prices.db (SQLite) ⟶ LightGBM Quantiles ⟶ CQR + ACI ⟶ Decision Engine ⟶ Trader PWA

3. Data Account, Temporality, Gaps & Quality Verification

Source	Time Span	Volume	Role, Temporality & Handling of Gaps
HARTI Daily Bulletin	2015–2026	3,097 parsed	Primary wholesale buy (P_t^w) and terminal sell (P_{t+1}^r) prices. 376,152 price cells. Dashes ("–") strictly stored as **NULL** (never zero). Pettah (pre-2021) & Peliyagoda unified.
CBSL Daily Price Report	2016–2026	2,415 parsed	Secondary cross-validation ($r = 0.951$–0.988), qualitative supply NLP reasons, and automated fallback when HARTI is delayed (scaled by 60d median ratio).
Open-Meteo Weather API	2015–2026	4 Agro-stations	3d, 7d, 14d rainfall sums for Nuwara Eliya, Welimada, Bandarawela, Dambulla (4,182d complete).
Macro & Calendar	2015–2026	99 Revisions	CPC auto-diesel revisions (C_t), CBSL USD/LKR exchange rate, astronomical Poya lunar cycle.

Time Handling & Integrity Audits (src/data_quality.py): Strict 06:00 AM cutoff enforced; walk-forward splits assert train $<$ cal $<$ test. Gaps are explicitly reported, never forward-filled: 39 missing bulletins at source (Feb 2017 404s); 94-day COVID lockdown in 2020 (a day-gap > 4 days automatically triggers **RED**). Zero inverted (low $>$ high) prices; 0 zero-price records; 390 core-corridor jumps $> 60\%$ verified and kept.

4. Feature Ablation: Empirical Value of Multi-Source Inputs

Evaluated via walk-forward testing on 2020–2026 (**results/ablation_features.json**):

- **Macro, Calendar & Extra Lags (2/14):** Narrows prediction intervals by up to 3.0 percentage points and reduces median error by up to Rs. 2.7/kg across all crops.
- **Agro-Rainfall Accumulations:** Demonstrates strong localized predictive power for perishable highland crops (narrows Tomato intervals to 59.1% and Leeks to 61.7%), while exhibiting neutral impact on sturdy lowland crops (Beans, Carrot).

5. Honest Evaluation Against Proposal on Real Data (2020–2026)

All metrics represent a walk-forward out-of-sample replay across **1,971 trading days per crop (1 Jan 2020 to 8 Nov 2026; 3,019 total corridor days)**. Decision thresholds were **strictly frozen on 2016–2019 data only** (**results/tuning_grid.csv**) with zero test snooping.

5.1 Interval Calibration: Is the 90% Prediction Range Honest?

Forecasting Method	Beans	Carrot	Tomato	Leeks	Green Chilli	Avg Width (% Price)
Naive (90-day historical ratio)	86.3%	88.0%	86.1%	86.8%	86.5%	54–71%
ARIMA(1,1,1)	90.6%	90.2%	90.1%	90.4%	90.1%	63–79% (Too wide)
Raw LightGBM Quantiles	78.1% **(Fail)**	77.6% **(Fail)**	78.3% **(Fail)**	75.7% **(Fail)**	77.0% **(Fail)**	39–48% (Undercovered)
Static Rolling CQR ($\gamma = 0$)	89.9%	89.4%	90.3%	89.8%	89.5%	55–70%
Model Y: CQR + ACI ($\gamma = 0.02$)	**90.0%**	**89.9%**	**90.2%**	**90.0%**	**90.0%**	**56–72% (Calibrated)**

Crisis Robustness: During Sri Lanka's 2022 economic collapse (fuel rationing, 80% inflation), uncalibrated LightGBM coverage plummeted to 79.5%–83.5%. ACI dynamically adapted non-conformity scores, maintaining **89.9%–91.4% honest coverage** throughout the crisis. Point forecast median error beats naive baselines on every crop (e.g., Beans median error Rs. 47/kg vs Rs. 57/kg for yesterday's Colombo quote).

5.2 Downside Protection: Catastrophic Loss Elimination (100 kg/day, $r = 0.3\%$/trip, $C =$ Rs. 6/kg)

A "Ruinous Day" is defined as a trading loss exceeding 30% of total capital deployed on that trip:

Vegetable Corridor	Always Trade P&L	Spread Rule P&L	Model Y Net P&L	Ruinous Days ($> 30\%$ loss) [Always / Spread / Model Y]	Worst Single Day Loss [Always / Model Y]
Beans	Rs. 5,075,938	Rs. 4,494,858	Rs. 2,964,791	43 / 28 / **21**	-Rs. 40,848 / **-Rs. 12,905**
Carrot	**-Rs. 1,404,118**	Rs. 89,319	**+Rs. 318,952**	64 / 19 / **2**	-Rs. 76,155 / **-Rs. 9,926**
Tomato	Rs. 6,153,725	Rs. 5,501,877	Rs. 3,226,784	17 / 13 / **14**	-Rs. 15,780 / **-Rs. 7,890**
Leeks	**-Rs. 1,692,692**	-Rs. 200,748	**+Rs. 102,888**	121 / 35 / **0**	-Rs. 68,385 / **-Rs. 3,086**
Green Chilli	Rs. 1,597,679	Rs. 1,994,036	Rs. 1,325,130	80 / 33 / **16**	-Rs. 45,960 / **-Rs. 21,657**
All 5 Corridors	**Rs. 9,730,532**	**Rs. 11,879,342**	**Rs. 7,938,545**	**325 / 128 / 53 (-84%)**	**Losses Capped by 50–95%**

Honest Analysis of the Economic Trade-Off:

1. **Solvency Protection on Failing Routes:** Model Y is the **only strategy profitable on all five corridors**. For chronic loss-making routes (Carrot and Leeks, where unconstrained trading lost Rs. 1.40M and Rs. 1.69M), Model Y transformed net losses into positive P&L (+Rs. 319k and +Rs. 103k).
2. **Massive Reduction in Catastrophic Wipeouts:** Model Y slashed ruinous loss days from 325 to **53 (an 84% elimination)**, completely zeroing wipeouts for Leeks (121 $\rightarrow$ 0) and cutting worst single-day losses by **50% to 95%**.
3. **Transparent Cost of Safety:** Model Y produces less total rupee profit than the naive spread rule (Rs. 7.94M vs Rs. 11.88M) because it commands AMBER (half quota) on uncertain days and issues RED on 52.5% of days. This is economically rational: traders on RED days experienced a 59%–67% empirical loss rate. For Beans alone, refusing RED trades forgave Rs. 0.91M in upside to avert **Rs. 1.57M in actual cash losses**.

5.3 System Usability: Market-Risk Index for Lenders & NGOs

We engineered an aggregate 0–100 Market-Risk Index (**results/risk_index.json**) for microfinance institutions. Stratifying 2020–2026 into five equal risk quintiles, the empirical probability of trade failure escalates monotonically:

Index Range: 17–40 (33.6% loss) $\longrightarrow$ 40–44 (40.5% loss) $\longrightarrow$ 44–48 (45.0% loss) $\longrightarrow$ 48–52 (51.0% loss) $\longrightarrow$ **52–74 (57.7% loss)** ($r = 0.36$)

6. Plan vs. Delivered Deliverables & Submission Artifacts

Proposal Specification	Delivered Status	Implementation Details & Execution Notes
Data Scraping & Store	**Complete**	3,097 HARTI + 2,415 CBSL bulletins in SQLite **prices.db**; Open-Meteo rain, diesel, FX.
Uncertainty Engine	**Complete**	LightGBM pinball quantiles + CQR + ACI ($\gamma = 0.02$). Verified with MAPIE (0.0 diff).
Downside Decision Logic	**Complete**	3-Tier engine ($B_t = P_t^w(1+r) + C$); forecast hidden on RED to stop anchoring.
Trader Delivery UI	**Complete**	Dual frontend: PWA on GitHub Pages (offline-ready) + Streamlit multi-tab analyst view.
Contingency Cuts (§8)	**Executed**	Omitted TFT deep model (prioritized tabular LightGBM); omitted SMS gateway (PWA used).

Official Deliverable Links:

- **Public Git Repository:** https://github.com/Jannahan/modelx
- **Live PWA Application:** https://jannahan.github.io/modelx/ui/
- **Team Model Y Sign-Off:** Tharusha Liyanagama · Sethuli Tinara Harischandra · Janakan Jude | IIT · ModelX 2.0